Esport Management System

Code:

```
test> use EsportManagement_FA2
switched to db EsportManagement_FA2
```

1)Coaches

```
EsportManagement_FA2> db.createCollection("coaches")
{ ok: 1 }
EsportManagement_FA2> db.coaches.insertMany([
|   {
|     coach_id: 1,
|     coach_name: "Vikram Mehta",
|     email: "vikram@novadragons.com",
|     experience_years: 5,
|     nationality: "India"
|   },
|   {
|     coach_id: 2,
|     coach_name: "James Carter",
|     email: "james@stormunit.com",
|     experience_years: 7,
|     nationality: "USA"
|   },
|   {
|     coach_id: 3,
|     coach_name: "Li Xiang",
|     email: "li@phantomx.com",
|     experience_years: 4,
|     nationality: "China"
|   },
|   {
|     coach_id: 4,
|     coach_name: "Erik Müller",
|     email: "erik@ironwolves.com",
|     experience_years: 6,
|     nationality: "Germany"
|   }
| ]);
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddbcd919103d366c3682d1'),
    '1': ObjectId('69ddbcd919103d366c3682d2'),
    '2': ObjectId('69ddbcd919103d366c3682d3'),
    '3': ObjectId('69ddbcd919103d366c3682d4')
  }
}
EsportManagement_FA2> db.coaches.find()
[
  {
    _id: ObjectId('69ddbcd919103d366c3682d1'),
    coach_id: 1,
    coach_name: 'Vikram Mehta',
    email: 'vikram@novadragons.com',
    experience_years: 5,
    nationality: 'India'
  },
  {
    _id: ObjectId('69ddbcd919103d366c3682d2'),
    coach_id: 2,
    coach_name: 'James Carter',
    email: 'james@stormunit.com',
    experience_years: 7,
    nationality: 'USA'
  },
  {
    _id: ObjectId('69ddbcd919103d366c3682d3'),
    coach_id: 3,
    coach_name: 'Li Xiang',
    email: 'li@phantomx.com',
    experience_years: 4,
    nationality: 'China'
  },
  {
    _id: ObjectId('69ddbcd919103d366c3682d4'),
    coach_id: 4,
    coach_name: 'Erik Müller',
    email: 'erik@ironwolves.com',
```

```
    experience_years: 6,
    nationality: 'Germany'
  }
]
```

2.Teams

```
EsportManagement_FA2> db.createCollection("teams");
{ ok: 1 }
EsportManagement_FA2> db.teams.insertMany([
|   {
|     team_id: 1,
|     team_name: "Nova Dragons",
|     region: "South Asia",
|     ranking: 1,
|     founded_year: 2020,
|     coach_id: 1
|   },
|   {
|     team_id: 2,
|     team_name: "Storm Unit",
|     region: "North America",
|     ranking: 2,
|     founded_year: 2019,
|     coach_id: 2
|   },
|   {
|     team_id: 3,
|     team_name: "Phantom X",
|     region: "East Asia",
|     ranking: 3,
|     founded_year: 2021,
|     coach_id: 3
|   },
|   {
|     team_id: 4,
|     team_name: "Iron Wolves",
|     region: "Europe",
|     ranking: 4,
|     founded_year: 2018,
|     coach_id: 4
```

```
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddbf7019103d366c3682d5'),
    '1': ObjectId('69ddbf7019103d366c3682d6'),
    '2': ObjectId('69ddbf7019103d366c3682d7'),
    '3': ObjectId('69ddbf7019103d366c3682d8')
  }
}
EsportManagement_FA2> db.teams.find()
[
  {
    _id: ObjectId('69ddbf7019103d366c3682d5'),
    team_id: 1,
    team_name: 'Nova Dragons',
    region: 'South Asia',
    ranking: 1,
    founded_year: 2020,
    coach_id: 1
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d6'),
    team_id: 2,
    team_name: 'Storm Unit',
    region: 'North America',
    ranking: 2,
    founded_year: 2019,
    coach_id: 2
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d7'),
    team_id: 3,
    team_name: 'Phantom X',
    region: 'East Asia',
    ranking: 3,
    founded_year: 2021,
    coach_id: 3
  },
```

```
  {
    _id: ObjectId('69ddbf7019103d366c3682d8'),
    team_id: 4,
    team_name: 'Iron Wolves',
    region: 'Europe',
    ranking: 4,
    founded_year: 2018,
    coach_id: 4
  }
]
```

3.Players Collection

```
EsportManagement_FA2> db.createCollection("players")
{ ok: 1 }
EsportManagement_FA2> db.players.insertMany([
|   {
|     player_id: 1,
|     player_name: "Alex Johnson",
|     gamertag: "xArcane",
|     age: 22,
|     nationality: "India",
|     role: "Duelist",
|     team_id: 1
|   },
|   {
|     player_id: 2,
|     player_name: "Rahul Sharma",
|     gamertag: "ShadowX",
|     age: 20,
|     nationality: "India",
|     role: "Controller",
|     team_id: 1
|   },
|   {
|     player_id: 3,
|     player_name: "Arjun Patel",
|     gamertag: "BladeStorm",
|     age: 24,
|     nationality: "India",
|     role: "Sentinel",
```

```
|     team_id: 1
|   },
|   {
|     player_id: 4,
|     player_name: "Chen Wei",
|     gamertag: "Z3nith",
|     age: 25,
|     nationality: "China",
|     role: "IGL",
|     team_id: 2
|   },
|   {
|     player_id: 5,
|     player_name: "Marcus Lee",
|     gamertag: "Reaper_V",
|     age: 19,
|     nationality: "USA",
|     role: "AWPer",
|     team_id: 2
|   },
|   {
|     player_id: 6,
|     player_name: "Tyler Brooks",
|     gamertag: "NightOwl",
|     age: 21,
|     nationality: "USA",
|     role: "Entry",
|     team_id: 2
|   },
|   {
EsportManagement_FA2> db.players.insertMany([
|   {
|     player_id: 1,
|     player_name: "Alex Johnson",
|     gamertag: "xArcane",
|     age: 22,
|     nationality: "India",
|     role: "Duelist",
|     team_id: 1
|   },
```

```
|   {
|     player_id: 2,
|     player_name: "Rahul Sharma",
|     gamertag: "ShadowX",
|     age: 20,
|     nationality: "India",
|     role: "Controller",
|     team_id: 1
|   },
|   {
|     player_id: 3,
|     player_name: "Arjun Patel",
|     gamertag: "BladeStorm",
|     age: 24,
|     nationality: "India",
|     role: "Sentinel",
|     team_id: 1
|   },
|   {
|     player_id: 4,
|     player_name: "Chen Wei",
|     gamertag: "Z3nith",
|     age: 25,
|     nationality: "China",
|     role: "IGL",
|     team_id: 2
|   },
|   {
|     player_id: 5,
|     player_name: "Marcus Lee",
|     gamertag: "Reaper_V",
|     age: 19,
|     nationality: "USA",
|     role: "AWPer",
|     team_id: 2
|   },
|   {
|     player_id: 6,
|     player_name: "Tyler Brooks",
|     gamertag: "NightOwl",
```

| age: 21,
| nationality: "USA",
| role: "Entry",
| team_id: 2
| },
| {
| player_id: 7,
| player_name: "Priya Nair",
| gamertag: "KryoN",
| age: 23,
| nationality: "India",
| role: "Duelist",
| team_id: 3
| },
| {
| player_id: 8,
| player_name: "Zhang Fei",
| gamertag: "PhantomBlade",
| age: 22,
| nationality: "China",
| role: "Lurker",
| team_id: 3
| },
| {
| player_id: 9,
| player_name: "Erik Bauer",
| gamertag: "IronFang",
| age: 26,
| nationality: "Germany",
| role: "IGL",
| team_id: 4
| },
| {
| player_id: 10,
| player_name: "Sofia Rossi",
| gamertag: "ViperX",
| age: 20,
| nationality: "Italy",
| role: "Controller",
| team_id: 4

```
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc01819103d366c3682d9'),
    '1': ObjectId('69ddc01819103d366c3682da'),
    '2': ObjectId('69ddc01819103d366c3682db'),
    '3': ObjectId('69ddc01819103d366c3682dc'),
    '4': ObjectId('69ddc01819103d366c3682dd'),
    '5': ObjectId('69ddc01819103d366c3682de'),
    '6': ObjectId('69ddc01819103d366c3682df'),
    '7': ObjectId('69ddc01819103d366c3682e0'),
    '8': ObjectId('69ddc01819103d366c3682e1'),
    '9': ObjectId('69ddc01819103d366c3682e2')
  }
}
EsportManagement_FA2> db.players.find()
[
  {
    _id: ObjectId('69ddc01819103d366c3682d9'),
    player_id: 1,
    player_name: 'Alex Johnson',
    gamertag: 'xArcane',
    age: 22,
    nationality: 'India',
    role: 'Duelist',
    team_id: 1
  },
  {
    _id: ObjectId('69ddc01819103d366c3682da'),
    player_id: 2,
    player_name: 'Rahul Sharma',
    gamertag: 'ShadowX',
    age: 20,
    nationality: 'India',
    role: 'Controller',
    team_id: 1
  },
  {
```

```
  _id: ObjectId('69ddc01819103d366c3682db'),
  player_id: 3,
  player_name: 'Arjun Patel',
  gamertag: 'BladeStorm',
  age: 24,
  nationality: 'India',
  role: 'Sentinel',
  team_id: 1
},
{
  _id: ObjectId('69ddc01819103d366c3682dc'),
  player_id: 4,
  player_name: 'Chen Wei',
  gamertag: 'Z3nith',
  age: 25,
  nationality: 'China',
  role: 'IGL',
  team_id: 2
},
{
  _id: ObjectId('69ddc01819103d366c3682dd'),
  player_id: 5,
  player_name: 'Marcus Lee',
  gamertag: 'Reaper_V',
  age: 19,
  nationality: 'USA',
  role: 'AWPer',
  team_id: 2
},
{
  _id: ObjectId('69ddc01819103d366c3682de'),
  player_id: 6,
  player_name: 'Tyler Brooks',
  gamertag: 'NightOwl',
  age: 21,
  nationality: 'USA',
  role: 'Entry',
  team_id: 2
},
{
```

```
    _id: ObjectId('69ddc01819103d366c3682df'),
    player_id: 7,
    player_name: 'Priya Nair',
    gamertag: 'KryoN',
    age: 23,
    nationality: 'India',
    role: 'Duelist',
    team_id: 3
  },
  {
    _id: ObjectId('69ddc01819103d366c3682e0'),
    player_id: 8,
    player_name: 'Zhang Fei',
    gamertag: 'PhantomBlade',
    age: 22,
    nationality: 'China',
    role: 'Lurker',
    team_id: 3
  },
  {
    _id: ObjectId('69ddc01819103d366c3682e1'),
    player_id: 9,
    player_name: 'Erik Bauer',
    gamertag: 'IronFang',
    age: 26,
    nationality: 'Germany',
    role: 'IGL',
    team_id: 4
  },
  {
    _id: ObjectId('69ddc01819103d366c3682e2'),
    player_id: 10,
    player_name: 'Sofia Rossi',
    gamertag: 'ViperX',
    age: 20,
    nationality: 'Italy',
    role: 'Controller',
    team_id: 4
  }
]
```

4.Venues Collection

```
EsportManagement_FA2> db.createCollection("venues")
{ ok: 1 }
EsportManagement_FA2> db.venues.insertMany([
|   {
|     venue_id: 1,
|     venue_name: "Esports Arena Mumbai",
|     city: "Mumbai",
|     capacity: 500
|   },
|   {
|     venue_id: 2,
|     venue_name: "Delhi Gaming Hub",
|     city: "Delhi",
|     capacity: 300
|   },
|   {
|     venue_id: 3,
|     venue_name: "Bangalore Tech Stadium",
|     city: "Bangalore",
|     capacity: 200
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc09019103d366c3682e3'),
    '1': ObjectId('69ddc09019103d366c3682e4'),
    '2': ObjectId('69ddc09019103d366c3682e5')
  }
}
EsportManagement_FA2> db.venues.find()
[
  {
    _id: ObjectId('69ddc09019103d366c3682e3'),
    venue_id: 1,
    venue_name: 'Esports Arena Mumbai',
    city: 'Mumbai',
    capacity: 500
  },
```

```
  {
    _id: ObjectId('69ddc09019103d366c3682e4'),
    venue_id: 2,
    venue_name: 'Delhi Gaming Hub',
    city: 'Delhi',
    capacity: 300
  },
  {
    _id: ObjectId('69ddc09019103d366c3682e5'),
    venue_id: 3,
    venue_name: 'Bangalore Tech Stadium',
    city: 'Bangalore',
    capacity: 200
  }
]
```

5.Tournaments Collection

```
EsportManagement_FA2> db.createCollection("tournaments")
{ ok: 1 }
EsportManagement_FA2> db.tournaments.insertMany([
|   {
|     tournament_id: 1,
|     tournament_name: "VALORANT Champions 2024",
|     start_date: "2024-03-01",
|     end_date: "2024-03-15",
|     game: "VALORANT",
|     prize_amount: 500000
|   },
|   {
|     tournament_id: 2,
|     tournament_name: "CS2 World Series",
|     start_date: "2024-04-10",
|     end_date: "2024-04-20",
|     game: "CS2",
|     prize_amount: 300000
|   },
|   {
|     tournament_id: 3,
|     tournament_name: "League of Legends Cup",
|     start_date: "2024-05-05",
```

```
|     end_date: "2024-05-18",
|     game: "League of Legends",
|     prize_amount: 450000
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc11a19103d366c3682e6'),
    '1': ObjectId('69ddc11a19103d366c3682e7'),
    '2': ObjectId('69ddc11a19103d366c3682e8')
  }
}
EsportManagement_FA2> db.tournaments.find()
[
  {
    _id: ObjectId('69ddc11a19103d366c3682e6'),
    tournament_id: 1,
    tournament_name: 'VALORANT Champions 2024',
    start_date: '2024-03-01',
    end_date: '2024-03-15',
    game: 'VALORANT',
    prize_amount: 500000
  },
  {
    _id: ObjectId('69ddc11a19103d366c3682e7'),
    tournament_id: 2,
    tournament_name: 'CS2 World Series',
    start_date: '2024-04-10',
    end_date: '2024-04-20',
    game: 'CS2',
    prize_amount: 300000
  },
  {
    _id: ObjectId('69ddc11a19103d366c3682e8'),
    tournament_id: 3,
    tournament_name: 'League of Legends Cup',
    start_date: '2024-05-05',
    end_date: '2024-05-18',
    game: 'League of Legends',
```

```
    prize_amount: 450000
  }
]
```

6)Matches Collection

```
EsportManagement_FA2> db.createCollection("matches")
{ ok: 1 }
EsportManagement_FA2> db.matches.insertMany([
|   {
|     match_id: 1,
|     tournament_id: 1,
|     venue_id: 1,
|     match_date: "2024-03-05",
EsportManagement_FA2> db.matches.insertMany([
|   {
|     match_id: 1,
|     tournament_id: 1,
|     venue_id: 1,
|     match_date: "2024-03-05",
|     team1_id: 1,
|     team2_id: 2,
|     seats_available: 20,
|     seats_total: 500
|   },
|   {
|     match_id: 2,
|     tournament_id: 1,
|     venue_id: 2,
|     match_date: "2024-03-08",
|     team1_id: 3,
|     team2_id: 4,
|     seats_available: 0,
|     seats_total: 300
|   },
|   {
|     match_id: 3,
|     tournament_id: 2,
|     venue_id: 1,
|     match_date: "2024-04-12",
|     team1_id: 1,
```

```
|     team2_id: 3,
|     seats_available: 50,
|     seats_total: 500
|   },
|   {
|     match_id: 4,
|     tournament_id: 3,
|     venue_id: 3,
|     match_date: "2024-05-10",
|     team1_id: 2,
|     team2_id: 4,
|     seats_available: 10,
|     seats_total: 200
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc15d19103d366c3682e9'),
    '1': ObjectId('69ddc15d19103d366c3682ea'),
    '2': ObjectId('69ddc15d19103d366c3682eb'),
    '3': ObjectId('69ddc15d19103d366c3682ec')
  }
}
EsportManagement_FA2> db.matches.find()
[
  {
    _id: ObjectId('69ddc15d19103d366c3682e9'),
    match_id: 1,
    tournament_id: 1,
    venue_id: 1,
    match_date: '2024-03-05',
    team1_id: 1,
    team2_id: 2,
    seats_available: 20,
    seats_total: 500
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682ea'),
    match_id: 2,
```

```
    tournament_id: 1,
    venue_id: 2,
    match_date: '2024-03-08',
    team1_id: 3,
    team2_id: 4,
    seats_available: 0,
    seats_total: 300
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682eb'),
    match_id: 3,
    tournament_id: 2,
    venue_id: 1,
    match_date: '2024-04-12',
    team1_id: 1,
    team2_id: 3,
    seats_available: 50,
    seats_total: 500
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682ec'),
    match_id: 4,
    tournament_id: 3,
    venue_id: 3,
    match_date: '2024-05-10',
    team1_id: 2,
    team2_id: 4,
    seats_available: 10,
    seats_total: 200
  }
]
```

7)Matchstats Collection

```
EsportManagement_FA2> db.createCollection("matchstats")
{ ok: 1 }
EsportManagement_FA2> db.matchstats.insertMany([
|   {
|     stats_id: 1,
|     player_id: 1,
|     match_id: 1,
```

| kills: 28,
| assists: 10,
| damage_done: 14500
| },
| {
| stats_id: 2,
| player_id: 2,
| match_id: 1,
| kills: 18,
| assists: 15,
| damage_done: 9800
| },
| {
| stats_id: 3,
| player_id: 3,
| match_id: 1,
| kills: 22,
| assists: 8,
| damage_done: 12200
| },
| {
| stats_id: 4,
| player_id: 4,
| match_id: 2,
| kills: 30,
| assists: 5,
| damage_done: 16800
| },
| {
| stats_id: 5,
| player_id: 5,
| match_id: 2,
| kills: 15,
| assists: 20,
| damage_done: 8500
| },
| {
| stats_id: 6,
| player_id: 6,
| match_id: 2,

```
|     kills: 20,
EsportManagement_FA2> db.matchstats.insertMany([
|   {
|     stats_id: 1,
|     player_id: 1,
|     match_id: 1,
|     kills: 28,
|     assists: 10,
|     damage_done: 14500
|   },
|   {
|     stats_id: 2,
|     player_id: 2,
|     match_id: 1,
|     kills: 18,
|     assists: 15,
|     damage_done: 9800
|   },
|   {
|     stats_id: 3,
|     player_id: 3,
|     match_id: 1,
|     kills: 22,
|     assists: 8,
|     damage_done: 12200
|   },
|   {
|     stats_id: 4,
|     player_id: 4,
|     match_id: 2,
|     kills: 30,
|     assists: 5,
|     damage_done: 16800
|   },
|   {
|     stats_id: 5,
|     player_id: 5,
|     match_id: 2,
|     kills: 15,
|     assists: 20,
```

| damage_done: 8500
| },
| {
| stats_id: 6,
| player_id: 6,
| match_id: 2,
| kills: 20,
| assists: 12,
| damage_done: 11000
| },
| {
| stats_id: 7,
| player_id: 7,
| match_id: 3,
| kills: 25,
| assists: 9,
| damage_done: 13400
| },
| {
| stats_id: 8,
| player_id: 8,
| match_id: 3,
| kills: 17,
| assists: 14,
| damage_done: 9200
| },
| {
| stats_id: 9,
| player_id: 9,
| match_id: 4,
| kills: 24,
| assists: 7,
| damage_done: 13000
| },
| {
| stats_id: 10,
| player_id: 10,
| match_id: 4,
| kills: 12,
| assists: 18,

```
|     damage_done: 7800
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc1a719103d366c3682ed'),
    '1': ObjectId('69ddc1a719103d366c3682ee'),
    '2': ObjectId('69ddc1a719103d366c3682ef'),
    '3': ObjectId('69ddc1a719103d366c3682f0'),
    '4': ObjectId('69ddc1a719103d366c3682f1'),
    '5': ObjectId('69ddc1a719103d366c3682f2'),
    '6': ObjectId('69ddc1a719103d366c3682f3'),
    '7': ObjectId('69ddc1a719103d366c3682f4'),
    '8': ObjectId('69ddc1a719103d366c3682f5'),
    '9': ObjectId('69ddc1a719103d366c3682f6')
  }
}
EsportManagement_FA2> db.matchstats.find()
[
  {
    _id: ObjectId('69ddc1a719103d366c3682ed'),
    stats_id: 1,
    player_id: 1,
    match_id: 1,
    kills: 28,
    assists: 10,
    damage_done: 14500
  },
  {
    _id: ObjectId('69ddc1a719103d366c3682ee'),
    stats_id: 2,
    player_id: 2,
    match_id: 1,
    kills: 18,
    assists: 15,
    damage_done: 9800
  },
  {
    _id: ObjectId('69ddc1a719103d366c3682ef'),
```

```
  stats_id: 3,
  player_id: 3,
  match_id: 1,
  kills: 22,
  assists: 8,
  damage_done: 12200
},
{
  _id: ObjectId('69ddc1a719103d366c3682f0'),
  stats_id: 4,
  player_id: 4,
  match_id: 2,
  kills: 30,
  assists: 5,
  damage_done: 16800
},
{
  _id: ObjectId('69ddc1a719103d366c3682f1'),
  stats_id: 5,
  player_id: 5,
  match_id: 2,
  kills: 15,
  assists: 20,
  damage_done: 8500
},
{
  _id: ObjectId('69ddc1a719103d366c3682f2'),
  stats_id: 6,
  player_id: 6,
  match_id: 2,
  kills: 20,
  assists: 12,
  damage_done: 11000
},
{
  _id: ObjectId('69ddc1a719103d366c3682f3'),
  stats_id: 7,
  player_id: 7,
  match_id: 3,
  kills: 25,
```

```
    assists: 9,
    damage_done: 13400
  },
  {
    _id: ObjectId('69ddc1a719103d366c3682f4'),
    stats_id: 8,
    player_id: 8,
    match_id: 3,
    kills: 17,
    assists: 14,
    damage_done: 9200
  },
  {
    _id: ObjectId('69ddc1a719103d366c3682f5'),
    stats_id: 9,
    player_id: 9,
    match_id: 4,
    kills: 24,
    assists: 7,
    damage_done: 13000
  },
  {
    _id: ObjectId('69ddc1a719103d366c3682f6'),
    stats_id: 10,
    player_id: 10,
    match_id: 4,
    kills: 12,
    assists: 18,
    damage_done: 7800
  }
]
```

8)Viewer Collection

```
EsportManagement_FA2> db.createCollection("viewers")
{ ok: 1 }
EsportManagement_FA2> db.viewers.insertMany([
|   {
|     viewer_id: 1,
|     viewer_name: "Rahul Gupta",
|     email: "rahul@gmail.com",
```

```
|     age: 21
|   },
|   {
|     viewer_id: 2,
|     viewer_name: "Priya Singh",
|     email: "priya@gmail.com",
|     age: 19
|   },
|   {
|     viewer_id: 3,
|     viewer_name: "Arjun Das",
|     email: "arjun@gmail.com",
|     age: 24
|   },
|   {
|     viewer_id: 4,
|     viewer_name: "Sneha Patil",
|     email: "sneha@gmail.com",
|     age: 22
|   },
|   {
|     viewer_id: 5,
|     viewer_name: "Karan Mehta",
|     email: "karan@gmail.com",
|     age: 20
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc1e719103d366c3682f7'),
    '1': ObjectId('69ddc1e719103d366c3682f8'),
    '2': ObjectId('69ddc1e719103d366c3682f9'),
    '3': ObjectId('69ddc1e719103d366c3682fa'),
    '4': ObjectId('69ddc1e719103d366c3682fb')
  }
}
EsportManagement_FA2> db.viewers.find()
[
  {
```

```
    _id: ObjectId('69ddc1e719103d366c3682f7'),
    viewer_id: 1,
    viewer_name: 'Rahul Gupta',
    email: 'rahul@gmail.com',
    age: 21
  },
  {
    _id: ObjectId('69ddc1e719103d366c3682f8'),
    viewer_id: 2,
    viewer_name: 'Priya Singh',
    email: 'priya@gmail.com',
    age: 19
  },
  {
    _id: ObjectId('69ddc1e719103d366c3682f9'),
    viewer_id: 3,
    viewer_name: 'Arjun Das',
    email: 'arjun@gmail.com',
    age: 24
  },
  {
    _id: ObjectId('69ddc1e719103d366c3682fa'),
    viewer_id: 4,
    viewer_name: 'Sneha Patil',
    email: 'sneha@gmail.com',
    age: 22
  },
  {
    _id: ObjectId('69ddc1e719103d366c3682fb'),
    viewer_id: 5,
    viewer_name: 'Karan Mehta',
    email: 'karan@gmail.com',
    age: 20
  }
]
```

9)Tickets Collection

```
EsportManagement_FA2> db.createCollection("tickets")
{ ok: 1 }
EsportManagement_FA2> db.tickets.insertMany([
EsportManagement_FA2> db.tickets.insertMany([
|   {
|     ticket_id: 1,
|     viewer_id: 1,
|     match_id: 1,
|     seat_no: "A12",
|     price: 500
|   },
|   {
|     ticket_id: 2,
|     viewer_id: 2,
|     match_id: 1,
|     seat_no: "A13",
|     price: 500
|   },
|   {
|     ticket_id: 3,
|     viewer_id: 3,
|     match_id: 2,
|     seat_no: "B05",
|     price: 300
|   },
|   {
|     ticket_id: 4,
|     viewer_id: 4,
|     match_id: 3,
|     seat_no: "C22",
|     price: 450
|   },
|   {
|     ticket_id: 5,
|     viewer_id: 5,
|     match_id: 4,
|     seat_no: "D08",
|     price: 350
|   }
```

```
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ddc23f19103d366c3682fc'),
    '1': ObjectId('69ddc23f19103d366c3682fd'),
    '2': ObjectId('69ddc23f19103d366c3682fe'),
    '3': ObjectId('69ddc23f19103d366c3682ff'),
    '4': ObjectId('69ddc23f19103d366c368300')
  }
}
EsportManagement_FA2> db.tickets.find()
[
  {
    _id: ObjectId('69ddc23f19103d366c3682fc'),
    ticket_id: 1,
    viewer_id: 1,
    match_id: 1,
    seat_no: 'A12',
    price: 500
  },
  {
    _id: ObjectId('69ddc23f19103d366c3682fd'),
    ticket_id: 2,
    viewer_id: 2,
    match_id: 1,
    seat_no: 'A13',
    price: 500
  },
  {
    _id: ObjectId('69ddc23f19103d366c3682fe'),
    ticket_id: 3,
    viewer_id: 3,
    match_id: 2,
    seat_no: 'B05',
    price: 300
  },
  {
    _id: ObjectId('69ddc23f19103d366c3682ff'),
    ticket_id: 4,
```

```
    viewer_id: 4,
    match_id: 3,
    seat_no: 'C22',
    price: 450
  },
  {
    _id: ObjectId('69ddc23f19103d366c368300'),
    ticket_id: 5,
    viewer_id: 5,
    match_id: 4,
    seat_no: 'D08',
    price: 350
  }
]
```

Queries

1)Register a New Player in the Esports Team Database

```
EsportManagement_FA2> db.players.insertOne({
|   player_id: 11,
|   player_name: "Rohan Verma",
|   gamertag: "StormRider",
|   age: 21,
|   nationality: "India",
|   role: "Duelist",
|   team_id: 2
| });
{
  acknowledged: true,
  insertedId: ObjectId('69ddc6b019103d366c368301')
}
```


```
EsportManagement_FA2> db.players.insertOne({
|   player_id: 11,
|   player_name: "Rohan Verma",
|   gamertag: "StormRider",
|   age: 21,
|   nationality: "India",
|   role: "Duelist",
|   team_id: 2
| });
{
  acknowledged: true,
  insertedId: ObjectId('69ddc6b019103d366c368301')
}
```


2)View Complete List of All Esports Teams

```
EsportManagement_FA2> db.teams.find().pretty()
[
  {
    _id: ObjectId('69ddbf7019103d366c3682d5'),
    team_id: 1,
    team_name: 'Nova Dragons',
    region: 'South Asia',
```

```
    ranking: 1,
    founded_year: 2020,
    coach_id: 1
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d6'),
    team_id: 2,
    team_name: 'Storm Unit',
    region: 'North America',
    ranking: 2,
    founded_year: 2019,
    coach_id: 2
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d7'),
    team_id: 3,
    team_name: 'Phantom X',
    region: 'East Asia',
    ranking: 3,
    founded_year: 2021,
    coach_id: 3
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d8'),
    team_id: 4,
    team_name: 'Iron Wolves',
    region: 'Europe',
    ranking: 4,
    founded_year: 2018,
    coach_id: 4
  }
]
```

```
EsportManagement_FA2> db.teams.find().pretty()
[
  {
    _id: ObjectId('69ddbf7019103d366c3682d5'),
    team_id: 1,
    team_name: 'Nova Dragons',
    region: 'South Asia',
    ranking: 1,
    founded_year: 2020,
    coach_id: 1
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d6'),
    team_id: 2,
    team_name: 'Storm Unit',
    region: 'North America',
    ranking: 2,
    founded_year: 2019,
    coach_id: 2
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d7'),
    team_id: 3,
    team_name: 'Phantom X',
    region: 'East Asia',
    ranking: 3,
    founded_year: 2021,
    coach_id: 3
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d8'),
    team_id: 4,
    team_name: 'Iron Wolves',
    region: 'Europe',
    ranking: 4,
    founded_year: 2018,
    coach_id: 4
```

3)Retrieve Players from a Specific Country (USA Players Analysis)

```
EsportManagement_FA2> db.players.find({nationality:"USA"})
[
  {
    _id: ObjectId('69ddc01819103d366c3682dd'),
    player_id: 5,
    player_name: 'Marcus Lee',
    gamertag: 'Reaper_V',
    age: 19,
```

```
    nationality: 'USA',
    role: 'AWPer',
    team_id: 2
  },
  {
    _id: ObjectId('69ddc01819103d366c3682de'),
    player_id: 6,
    player_name: 'Tyler Brooks',
    gamertag: 'NightOwl',
    age: 21,
    nationality: 'USA',
    role: 'Entry',
    team_id: 2
  }
]
```

```
EsportManagement_FA2> db.players.find({nationality:"USA"})
[
  {
    _id: ObjectId('69ddc01819103d366c3682dd'),
    player_id: 5,
    player_name: 'Marcus Lee',
    gamertag: 'Reaper_V',
    age: 19,
    nationality: 'USA',
    role: 'AWPer',
    team_id: 2
  },
  {
    _id: ObjectId('69ddc01819103d366c3682de'),
    player_id: 6,
    player_name: 'Tyler Brooks',
    gamertag: 'NightOwl',
    age: 21,
    nationality: 'USA',
    role: 'Entry',
    team_id: 2
  }
]
```

4)Display Team Rankings and Regions for Tournament Overview

```
EsportManagement_FA2> db.teams.find({},{team_name:1,region:1,ranking:1})
[
  {
    _id: ObjectId('69ddbf7019103d366c3682d5'),
    team_name: 'Nova Dragons',
    region: 'South Asia',
    ranking: 1
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d6'),
    team_name: 'Storm Unit',
    region: 'North America',
    ranking: 2
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d7'),
    team_name: 'Phantom X',
    region: 'East Asia',
    ranking: 3
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d8'),
    team_name: 'Iron Wolves',
    region: 'Europe',
    ranking: 4
  }
]
```

```
EsportManagement_FA2> db.teams.find({},{team_name:1,region:1,ranking:1})
[
  {
    _id: ObjectId('69ddbf7019103d366c3682d5'),
    team_name: 'Nova Dragons',
    region: 'South Asia',
    ranking: 1
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d6'),
    team_name: 'Storm Unit',
    region: 'North America',
    ranking: 2
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d7'),
    team_name: 'Phantom X',
    region: 'East Asia',
    ranking: 3
  },
  {
    _id: ObjectId('69ddbf7019103d366c3682d8'),
    team_name: 'Iron Wolves',
    region: 'Europe',
    ranking: 4
  }
]
```

5)Update Player Role Based on Performance or Strategy Change

EsportManagement_FA2>
db.players.updateOne({gamertag:"ShadowX"},{$set:{role:"IGL"}})

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
EsportManagement_FA2> db.players.updateOne({gamertag:"ShadowX"},{$set:{role:"IGL"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

6) Remove a Venue from the System (Closed or Unavailable)

EsportManagement_FA2> db.venues.deleteOne({venue_id:3})

{ acknowledged: true, deletedCount: 1 }

```
EsportManagement_FA2> db.venues.deleteOne({venue_id:3})
{ acknowledged: true, deletedCount: 1 }
```

7)Analyze Player Performance Across Matches

EsportManagement_FA2> db.matchstats.aggregate([

| {

```
|     $group:{
|       _id:"$player_id",
|       total_kills:{$sum:"$kills"},
|       total_assists:{$sum:"$assists"},
|       total_damage:{$sum:"$damage_done"}
|     }
|   },
|   {
|     $sort:{total_kills:-1}
|   }
| ])
[
  { _id: 4, total_kills: 30, total_assists: 5, total_damage: 16800 },
  { _id: 1, total_kills: 28, total_assists: 10, total_damage: 14500 },
  { _id: 7, total_kills: 25, total_assists: 9, total_damage: 13400 },
  { _id: 9, total_kills: 24, total_assists: 7, total_damage: 13000 },
  { _id: 3, total_kills: 22, total_assists: 8, total_damage: 12200 },
  { _id: 6, total_kills: 20, total_assists: 12, total_damage: 11000 },
  { _id: 2, total_kills: 18, total_assists: 15, total_damage: 9800 },
  { _id: 8, total_kills: 17, total_assists: 14, total_damage: 9200 },
  { _id: 5, total_kills: 15, total_assists: 20, total_damage: 8500 },
  { _id: 10, total_kills: 12, total_assists: 18, total_damage: 7800 }
]
```

```
EsportManagement_FA2> db.matchstats.aggregate([
|   {
|     $group:{
|       _id:"$player_id",
|       total_kills:{$sum:"$kills"},
|       total_assists:{$sum:"$assists"},
|       total_damage:{$sum:"$damage_done"}
|     }
|   },
|   {
|     $sort:{total_kills:-1}
|   }
| ])
[
  { _id: 4, total_kills: 30, total_assists: 5, total_damage: 16800 },
  { _id: 1, total_kills: 28, total_assists: 10, total_damage: 14500 },
  { _id: 7, total_kills: 25, total_assists: 9, total_damage: 13400 },
  { _id: 9, total_kills: 24, total_assists: 7, total_damage: 13000 },
  { _id: 3, total_kills: 22, total_assists: 8, total_damage: 12200 },
  { _id: 6, total_kills: 20, total_assists: 12, total_damage: 11000 },
  { _id: 2, total_kills: 18, total_assists: 15, total_damage: 9800 },
  { _id: 8, total_kills: 17, total_assists: 14, total_damage: 9200 },
  { _id: 5, total_kills: 15, total_assists: 20, total_damage: 8500 },
  { _id: 10, total_kills: 12, total_assists: 18, total_damage: 7800 }
]
```

8) Calculate Team Strength Based on Number of Players

```
EsportManagement_FA2> db.players.aggregate([
|   {
|     $group:{
|       _id:"$team_id",
|       total_players:{$sum:1}
|     }
|   },
|   {
|     $sort:{total_players:-1}
|   }
| ])
[
  { _id: 2, total_players: 4 },
  { _id: 1, total_players: 3 },
  { _id: 3, total_players: 2 },
  { _id: 4, total_players: 2 }
]
```

```
EsportManagement_FA2> db.players.aggregate([
|   {
|     $group:{
|       _id:"$team_id",
|       total_players:{$sum:1}
|     }
|   },
|   {
|     $sort:{total_players:-1}
|   }
| ])
[
  { _id: 2, total_players: 4 },
  { _id: 1, total_players: 3 },
  { _id: 3, total_players: 2 },
  { _id: 4, total_players: 2 }
]
```

9)Generate Ticket Sales and Revenue Report for Each Match

```
EsportManagement_FA2> db.tickets.aggregate([
|  {
|    $group:{
|      _id:"$match_id",
|      total_tickets_sold:{$sum:1},
|      total_revenue:{$sum:"$price"},
|      avg_ticket_price:{$avg:"$price"}
|    }
|  },
|  {
|    $sort:{total_revenue:-1}
|  },
|  {
|    $project:{
|      match_id:"$_id",
|      total_tickets_sold:1,
|      total_revenue:1,
|      avg_ticket_price:{$round:["$avg_ticket_price",2]},
|      _id:0
|    }
```

```
|   }
| ])
[
  {
    total_tickets_sold: 2,
    total_revenue: 1000,
    match_id: 1,
    avg_ticket_price: 500
  },
  {
    total_tickets_sold: 1,
    total_revenue: 450,
    match_id: 3,
    avg_ticket_price: 450
  },
  {
    total_tickets_sold: 1,
    total_revenue: 350,
    match_id: 4,
    avg_ticket_price: 350
  },
  {
    total_tickets_sold: 1,
    total_revenue: 300,
    match_id: 2,
    avg_ticket_price: 300
  }
]
```

```
EsportManagement_FA2> db.tickets.aggregate([
|   {
|     $group:{
|       _id:"$match_id",
|       total_tickets_sold:{$sum:1},
|       total_revenue:{$sum:"$price"},
|       avg_ticket_price:{$avg:"$price"}
|     }
|   },
|   {
|     $sort:{total_revenue:-1}
|   },
|   {
|     $project:{
|       match_id:"$_id",
|       total_tickets_sold:1,
|       total_revenue:1,
|       avg_ticket_price:{$round:["$avg_ticket_price",2]},
|       _id:0
|     }
|   }
| ])
[
  {
    total_tickets_sold: 2,
    total_revenue: 1000,
    match_id: 1,
    avg_ticket_price: 500
  },
  {
    total_tickets_sold: 1,
    total_revenue: 450,
    match_id: 3,
    avg_ticket_price: 450
  },
  {
    total_tickets_sold: 1,
    total_revenue: 350,
    match_id: 4,
    avg_ticket_price: 350
  },
  {
    total_tickets_sold: 1,
    total_revenue: 300,
    match_id: 2,
    avg_ticket_price: 300
  }
]
```

10) Identify Fully Booked Matches (No Seats Available)

```
EsportManagement_FA2> db.matches.find({
|   seats_available: 0
| })
```

```
[
  {
    _id: ObjectId('69ddc15d19103d366c3682ea'),
    match_id: 2,
    tournament_id: 1,
    venue_id: 2,
    match_date: '2024-03-08',
    team1_id: 3,
    team2_id: 4,
    seats_available: 0,
    seats_total: 300
  }
]
```

```
EsportManagement_FA2> db.matches.find({
|   seats_available: 0
| })
[
  {
    _id: ObjectId('69ddc15d19103d366c3682ea'),
    match_id: 2,
    tournament_id: 1,
    venue_id: 2,
    match_date: '2024-03-08',
    team1_id: 3,
    team2_id: 4,
    seats_available: 0,
    seats_total: 300
  }
]
EsportManagement_FA2> |
```

11)Optimize Player Search Using Indexing on Gamertag

```
EsportManagement_FA2> db.players.createIndex({ gamertag: 1 })
gamertag_1
EsportManagement_FA2> db.players.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { gamertag: 1 }, name: 'gamertag_1' }
]
EsportManagement_FA2> db.players.find({ gamertag: "ShadowX" })
```

```
[
  {
    _id: ObjectId('69ddc01819103d366c3682da'),
    player_id: 2,
    player_name: 'Rahul Sharma',
    gamertag: 'ShadowX',
    age: 20,
    nationality: 'India',
    role: 'IGL',
    team_id: 1
  }
]
```

```
EsportManagement_FA2> db.players.createIndex({ gamertag: 1 })
gamertag_1
EsportManagement_FA2> db.players.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { gamertag: 1 }, name: 'gamertag_1' }
]
EsportManagement_FA2> db.players.find({ gamertag: "ShadowX" })
[
  {
    _id: ObjectId('69ddc01819103d366c3682da'),
    player_id: 2,
    player_name: 'Rahul Sharma',
    gamertag: 'ShadowX',
    age: 20,
    nationality: 'India',
    role: 'IGL',
    team_id: 1
  }
]
EsportManagement_FA2>
```

12)Detect Matches with Low Seat Availability (High Demand Matches)

```
EsportManagement_FA2> db.matches.aggregate([
|  {
|    $project:{
|      match_id:1,
|      seats_available:1,
|      seats_total:1,
|      availability_ratio:{
|        $divide:["$seats_available","$seats_total"]
|      }
```

```
|     }
|   },
|   {
|     $match:{
|       availability_ratio:{$lt:0.1}
|     }
|   }
| ])
[
  {
    _id: ObjectId('69ddc15d19103d366c3682e9'),
    match_id: 1,
    seats_available: 20,
    seats_total: 500,
    availability_ratio: 0.04
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682ea'),
    match_id: 2,
    seats_available: 0,
    seats_total: 300,
    availability_ratio: 0
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682ec'),
    match_id: 4,
    seats_available: 10,
    seats_total: 200,
    availability_ratio: 0.05
  }
]
```

```
EsportManagement_FA2> db.matches.aggregate([
|   {
|     $project:{
|       match_id:1,
|       seats_available:1,
|       seats_total:1,
|       availability_ratio:{
|         $divide:["$seats_available","$seats_total"]
|       }
|     }
|   },
|   {
|     $match:{
|       availability_ratio:{$lt:0.1}
|     }
|   }
| ])
[
  {
    _id: ObjectId('69ddc15d19103d366c3682e9'),
    match_id: 1,
    seats_available: 20,
    seats_total: 500,
    availability_ratio: 0.04
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682ea'),
    match_id: 2,
    seats_available: 0,
    seats_total: 300,
    availability_ratio: 0
  },
  {
    _id: ObjectId('69ddc15d19103d366c3682ec'),
    match_id: 4,
    seats_available: 10,
    seats_total: 200,
    availability_ratio: 0.05
  }
]
```